

Appendix 4

All functional and non-functional requirements:

Software independence:

- **FR1:** The system shall produce a voter-verifiable paper audit trail (VVAT) for every cast vote, such that election outcomes can be audited independently of the software.
- **NFR1:** The paper audit trail shall remain the authoritative record of the election outcome; any discrepancy between the software tally and the physical record shall trigger an automatic audit halt.

End-to-end verifiability:

- **FR1:** The system shall provide each voter with a unique tracking code after casting their vote, which they can use to verify that their vote appears correctly in the published tally.
- **FR2:** The system shall publish a cryptographically signed bulletin board containing all cast ballots and the final tally, accessible to any third-party auditor.
- **NFR1:** The system's verifiability mechanisms shall function correctly regardless of the underlying software or hardware used to conduct the election, such that no undetected software error can produce an undetectable change in the verified outcome.

Collection Accountability

- **FR1:** The system shall generate a timestamped confirmation receipt for each submitted vote, which the voter can use as evidence of submission if their vote is not reflected in the tally.
- **NFR1:** The system shall retain submission logs for a minimum of 90 days post-election, accessible to electoral authorities for dispute investigation.

Dispute Resolution

- **FR1:** The system shall provide a third-party auditor interface through which any dispute between a voter and the election authority can be adjudicated using cryptographic proofs.
- **FR2:** The system shall automatically flag and log any ballot confirmation check that returns an invalid proof, making the record available to a neutral arbitrator.

- **NFR1:** Dispute resolution proofs shall be computationally verifiable by a third party within 60 seconds on standard consumer hardware.

Verifiability (Cast-as-Intended)

- **FR1:** After ballot marking but before final submission, the system shall display a human-readable summary of the voter's selections for explicit confirmation.
- **NFR1:** The cast-as-intended confirmation step shall be presented in a format accessible to voters with visual impairments, in compliance with WCAG 2.1 AA.

Verifiability (Recorded-as-Cast)

- **FR1:** The system shall allow each voter to query the bulletin board using their tracking code to confirm that their recorded ballot matches what they submitted.
- **NFR1:** The bulletin board query interface shall be available for at least 14 days following the close of the election.

Verifiability (Counted-as-Recorded)

- **FR1:** The system shall publish the complete set of recorded ballots and the final tally in a form that allows any member of the public to independently recompute and confirm the election result.
- **NFR1:** The published proof and tally data shall be made available in a machine-readable open format (e.g. JSON) within 24 hours of polls closing.

Verifiability (Other)

- **FR1:** The system shall ensure that the list of participating voters is publicly verifiable, allowing any observer to confirm the total number of votes cast without revealing the identity of individual voters.
- **NFR1:** All verifiability artifacts shall be archived and publicly accessible for a minimum of five years following the election.

Everlasting Privacy

- **FR1:** The system shall encrypt ballots using information-theoretically secure methods (e.g. secret sharing) that do not rely on the assumed hardness of any computational problem.
- **NFR1:** Ballot confidentiality shall remain guaranteed even in the event that all cryptographic primitives used in the system are broken at a future date.

Receipt-Freeness

- **FR1:** The system shall be designed such that a voter has no means of demonstrating to a third party which candidate or option they voted for after the vote has been cast.
- **FR2:** In polling station deployments, the system shall discard all intermediate ballot state after the vote is cast, leaving no exportable record of individual choices.
- **NFR1:** The system shall be designed such that any purported vote receipt produced by a voter is cryptographically indistinguishable from a forged one.

Coercion resistance

- **FR1:** The system shall ensure that regardless of the strategy employed by a coercer, a voter retains the ability to cast their intended vote in a way that cannot be detected or verified by the coercer.
- **NFR1:** The coercion-resistance mechanism shall not introduce any observable difference in system behavior or response time compared to normal voting.

Availability

- **FR1:** The system shall be deployed across geographically distributed servers such that the failure of any single node does not interrupt the voting service.
- **FR2:** The system shall implement rate limiting and traffic filtering to mitigate selective denial-of-service attacks targeting individual voter demographics.
- **NFR1:** The system shall maintain 99.9% uptime during the election window, with a maximum tolerated unplanned outage of 5 minutes.

Usability (Efficiency)

- **FR1:** The voting interface shall minimize the number of interactions required to complete ballot submission, removing any steps not strictly necessary to securely cast a vote.
- **NFR1:** The system shall not impose unnecessary latency between voter actions, ensuring that interface responsiveness does not become a barrier to timely vote completion.

Usability (Effectiveness)

- **FR1:** The system shall implement a ballot confirmation screen that prevents accidental submission of incomplete or unintended ballots.
- **NFR1:** Usability testing shall demonstrate that at least 95% of test participants are able to cast a correct ballot without assistance.

Usability (Satisfaction)

- **FR1:** The system shall provide voters with sufficient feedback at each stage of the voting process so that they feel informed and confident that their vote has been correctly received.
- **NFR1:** The system shall be designed to avoid voter confusion or frustration, at the submission and confirmation stages.

Usability (Transparency)

- **FR1:** The system shall publish plain-language documentation explaining how votes are recorded, encrypted, and tallied, accessible from within the voting interface.
- **NFR1:** All published transparency documentation shall be reviewed and approved by an independent electoral observer body before each election.

Usability (Other)

- **FR1:** The system shall provide an accessible voting mode compliant with assistive technologies (screen readers, switch access) for voters with disabilities.
- **NFR1:** Accessibility compliance shall be verified against WCAG 2.1 Level AA prior to each election deployment.

Auditability

- **FR1:** The system shall maintain a tamper-evident record of all significant system events throughout the election, sufficient to allow authorized auditors to reconstruct and verify the conduct of the election after the fact.
- **FR2:** Authorized auditors shall be able to export the full audit log in a standardized format for independent review.
- **NFR1:** The audit log shall be cryptographically signed at regular intervals not exceeding 10 minutes throughout the election period.

Correctness

- **FR1:** The tallying module shall apply a verifiable homomorphic computation or mix-net process to ensure all valid ballots are counted exactly once and no invalid ballots are included.
- **NFR1:** The system shall produce a correctness proof verifiable by an independent auditor within 2 hours of the polls closing.

Privacy

- **FR1:** The system shall encrypt all ballots at the point of entry using public-key encryption, such that no single authority can decrypt an individual ballot.

- **FR2:** Voter identity and ballot content shall be stored in separate, access-controlled databases with no way to query a join between them.
- **NFR1:** Privacy protections shall remain effective against a passive adversary observing all network traffic throughout and after the election.

Unreuseability

- **FR1:** The system shall mark each voter's credential as used immediately upon ballot submission, preventing any subsequent submission attempt using the same credential.
- **NFR1:** The system shall ensure that once a voter's credential has been used to cast a vote, it is permanently invalidated for the remainder of the election period, regardless of the channel or device through which a subsequent submission attempt is made.

Eligibility

- **FR1:** The system shall verify each voter's eligibility against the official electoral register before granting access to the ballot.
- **FR2:** In hybrid deployments, eligibility checks performed at physical polling stations shall be synchronized in real time with the central electoral register to prevent duplicate participation across channels.
- **NFR1:** The system shall complete eligibility verification within a response time that does not meaningfully delay the voter's access to the ballot, even when a large number of voters are authenticating simultaneously.

Robustness

- **FR1:** The system shall continue to accept and process votes correctly if up to one third of back-end nodes fail or behave maliciously.
- **NFR1:** Recovery from a partial system failure shall be automatic and shall not require manual intervention, restoring full service within 2 minutes.

Fairness

- **FR1:** The system shall ensure that no election authority or other party is able to determine any aspect of the election outcome before the voting period has officially ended.
- **NFR1:** The system shall be structured such that finalizing the tally requires the cooperation of multiple independent parties, preventing any single actor from unilaterally accessing results ahead of schedule.

Practicality

- **FR1:** The system shall be operable without requiring voters to install dedicated software, relying only on a standard web browser and internet connection.
- **FR2:** The cryptographic protocols used shall be configurable to scale from small municipal elections (hundreds of voters) to national elections (millions of voters) without architectural changes.
- **NFR1:** The system shall function within the network and hardware constraints typical of rural or low-bandwidth polling environments.

Safe Aggregation

- **FR1:** In hybrid elections, the system shall provide a dedicated aggregation module that accepts both electronically cast votes and scanned paper ballots, combining them into a single verifiable tally.
- **NFR1:** The aggregated tally shall be cryptographically demonstrable as the correct combination of both channels, verifiable by an independent auditor.

Authentication

- **FR1:** The system shall authenticate each voter using multi-factor authentication before granting access to the ballot.
- **NFR1:** Authentication credentials shall be invalidated immediately after use and shall not be reusable within the same election period.

Integrity

- **FR1:** All data transmitted between the voter's device and the voting server shall be protected by end-to-end encryption, preventing any intermediary from altering ballot content in transit.
- **NFR1:** Any detected tampering with stored ballot data shall trigger an automatic system halt.

Forward Secrecy

- **FR1:** The system shall ensure that the confidentiality of previously cast ballots is not compromised if cryptographic keys used during the election are exposed or stolen at a later point in time.
- **NFR1:** Session keys shall be rotated and securely deleted at the end of every individual voting session, with no recoverable copy retained.

Non-Valid Voting Capability

- **FR1:** The system shall allow voters to deliberately submit a blank or intentionally invalid ballot as a legitimate act, without this affecting the validity of other votes or the integrity of the tally.
- **NFR1:** Invalid votes shall be recorded and included in published election statistics as a separate category, independently of the valid vote tally.

Anonymity

- **FR1:** The system shall ensure that the link between a voter's identity and their ballot is permanently and irrevocably broken before the ballot is included in the tally.
- **NFR1:** No combination of system logs, network captures, or database records shall be sufficient to link a specific unsealed ballot to a specific voter.

Revocation

- **FR1:** In internet voting deployments, the system shall allow a voter to cancel and recast their vote at any point before the close of polls, with only the most recent valid submission counted.
- **NFR1:** Vote revocation shall be processed within 5 seconds, and the previous ballot shall be cryptographically invalidated with no possibility of recovery.